

Lecture 9

Design for Testability

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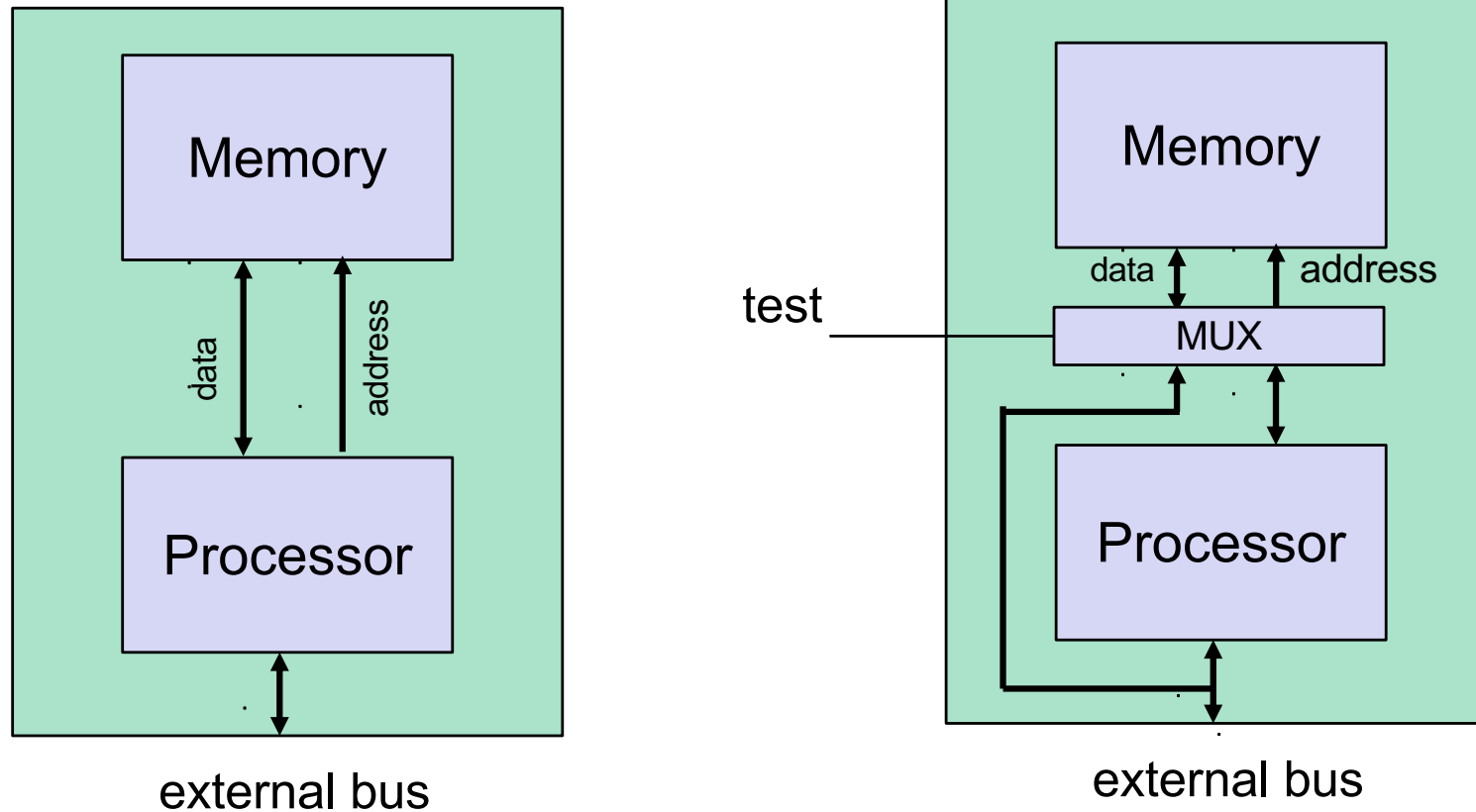
Test Approaches

1. Ad-hoc testing
2. Scan-based Test
3. Built-in Self-Test (BIST)

Problem is getting harder

- increasing complexity and heterogeneous combination of modules in system-on-a-chip.
- Advanced packaging and assembly techniques extend problem to the board level

Ad-hoc Test



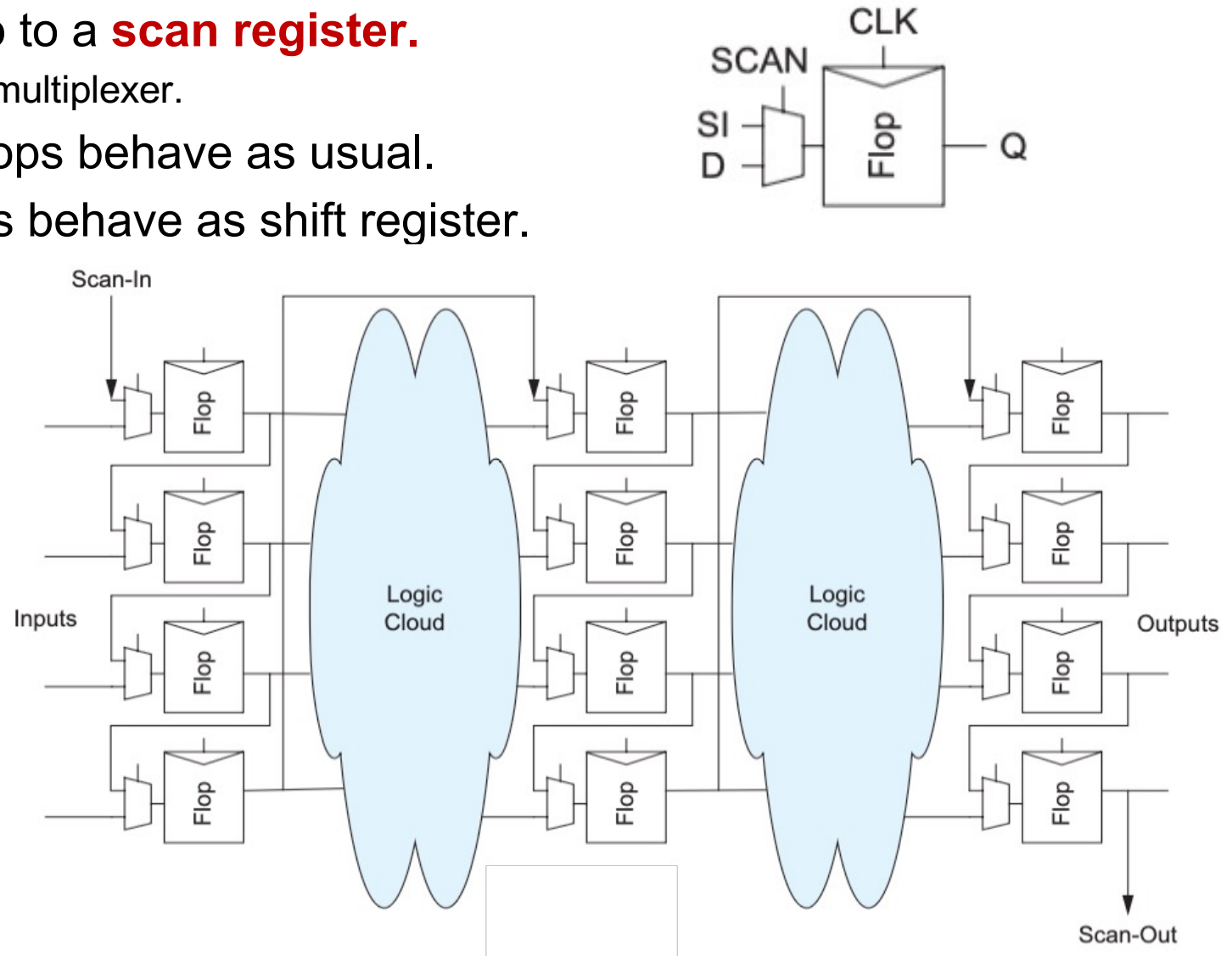
Inserting multiplexer improves testability

Design for Testability (DFT)

- ◆ Design the chip to increase observability and controllability
- ◆ If each register could be observed and controlled, test problem reduces to testing combinational logic between registers.
- ◆ Better yet, logic blocks could enter test mode where they generate test patterns and report the results automatically.

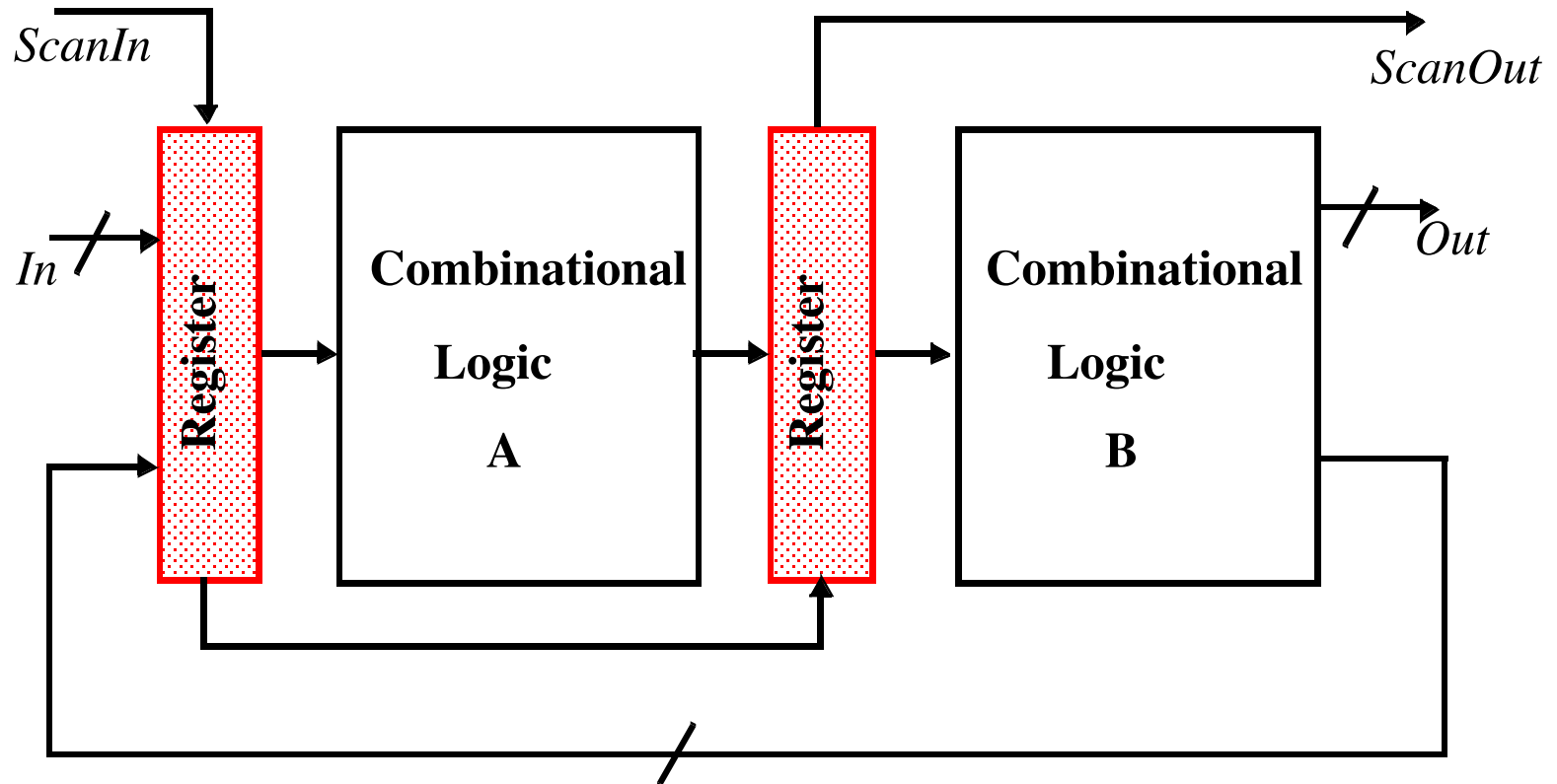
Scan Register

- ◆ Convert each flip-flop to a **scan register**.
 - Only costs one extra multiplexer.
- ◆ **Normal mode**: flip-flops behave as usual.
- ◆ **Scan mode**: flip-flops behave as shift register.
- ◆ Contents of flip-flops can be scanned out and new values scanned in.

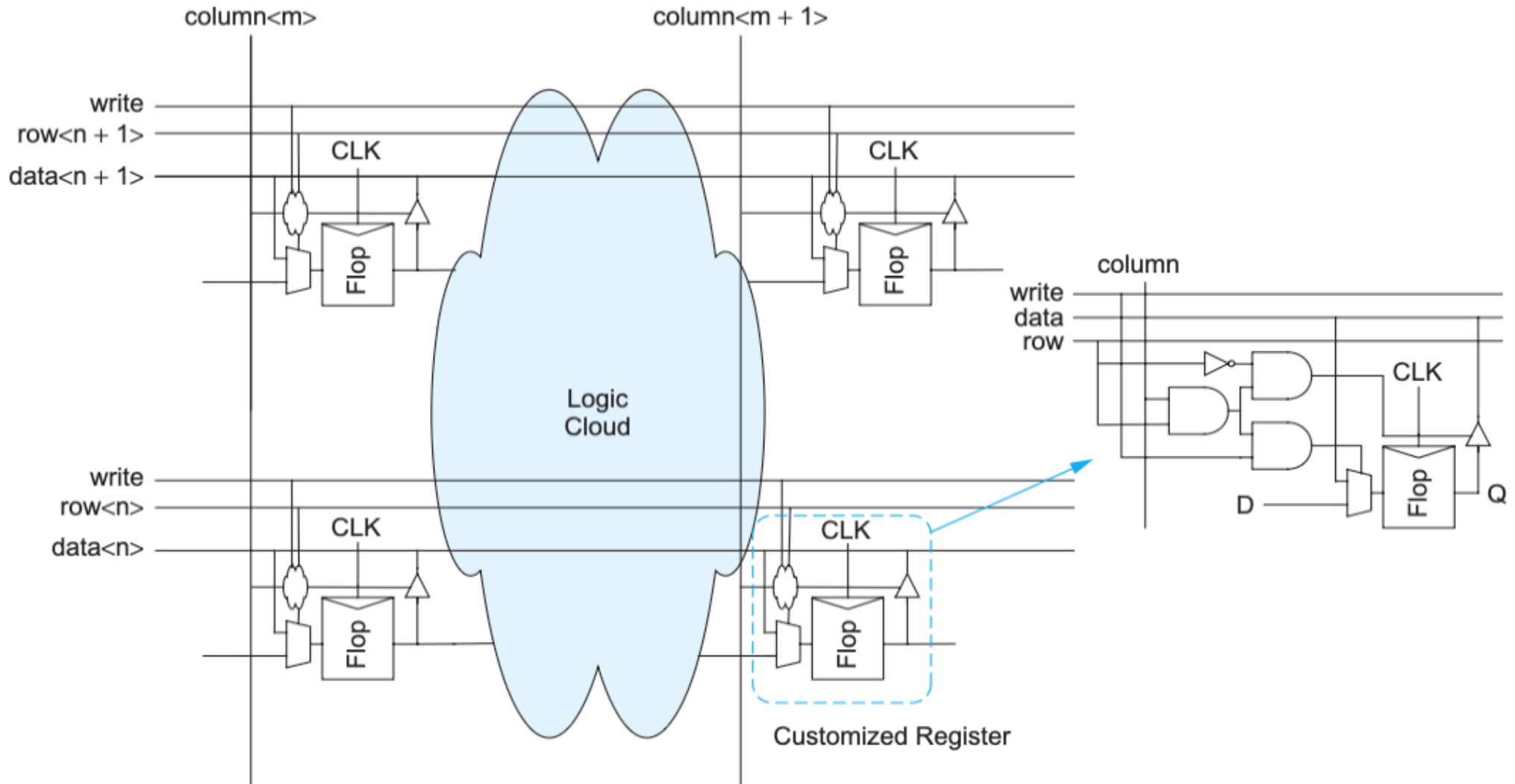


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Scan-based Test

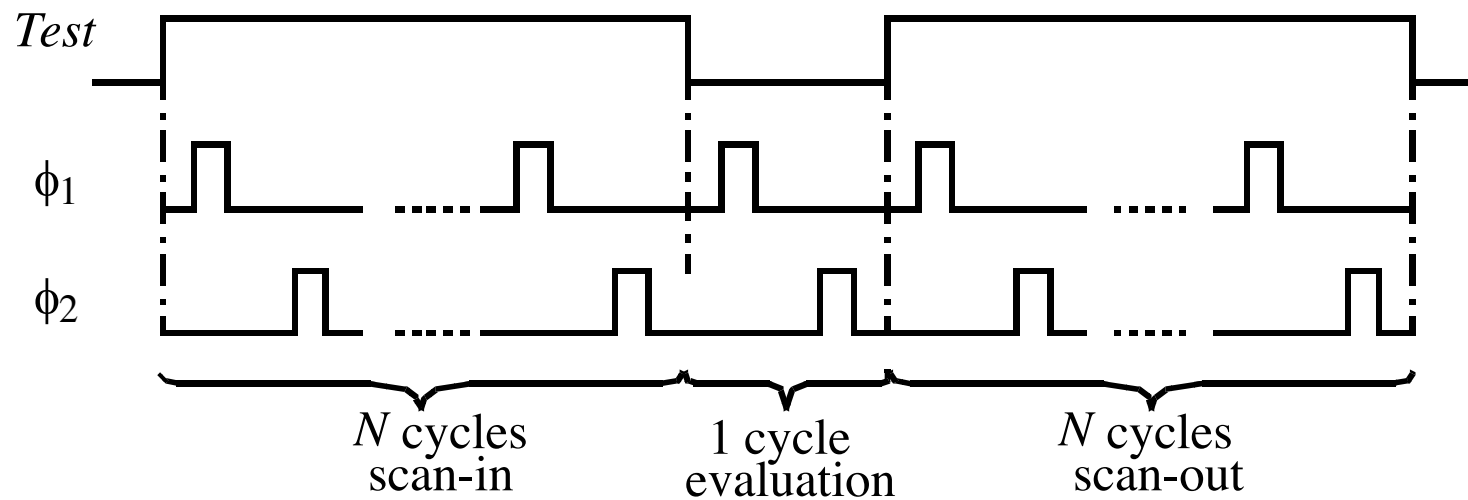
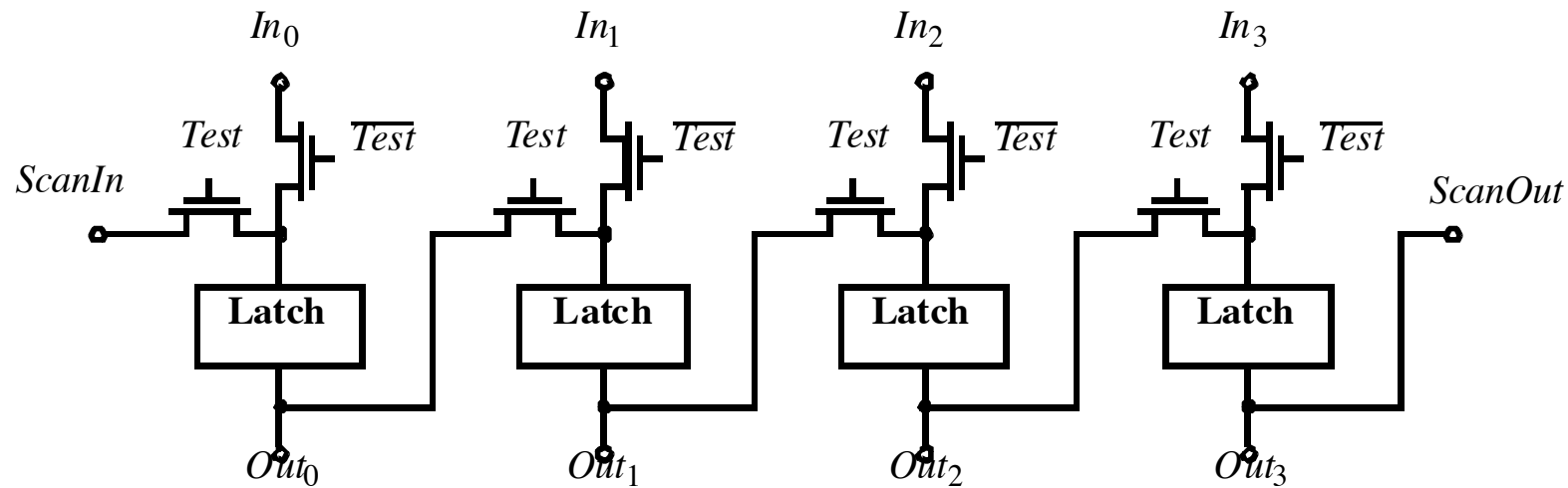


Parallel Scan Structure

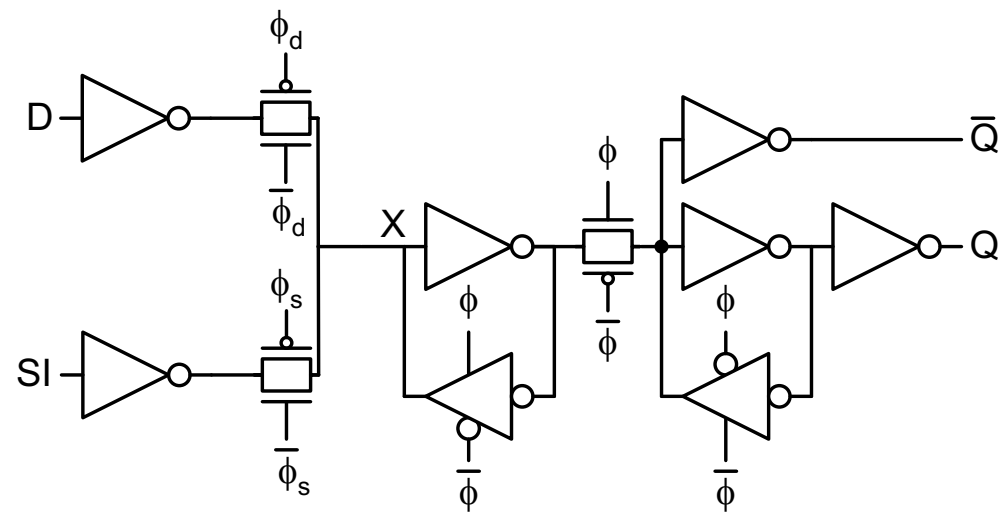
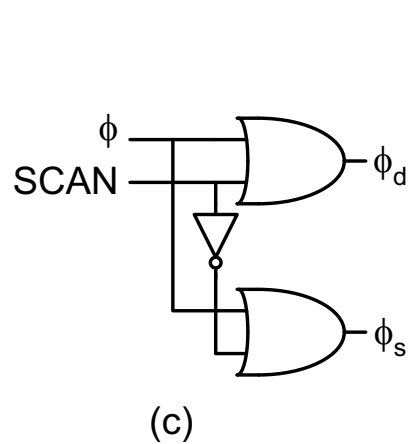
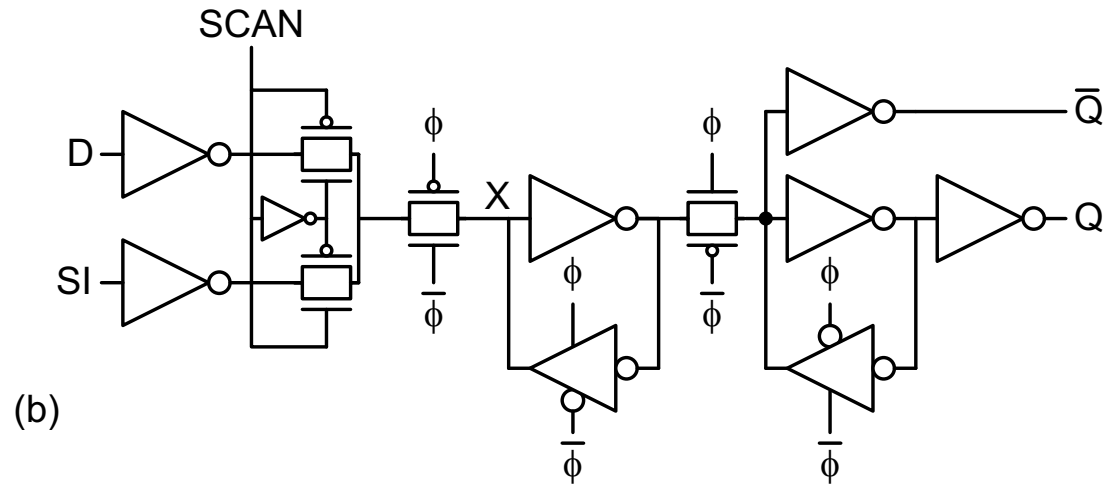
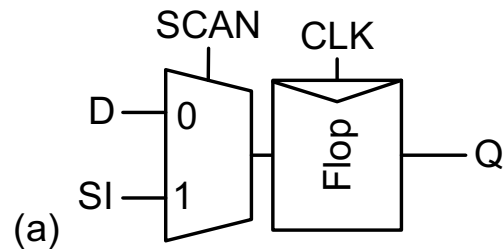


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Scan-based Test — Operation

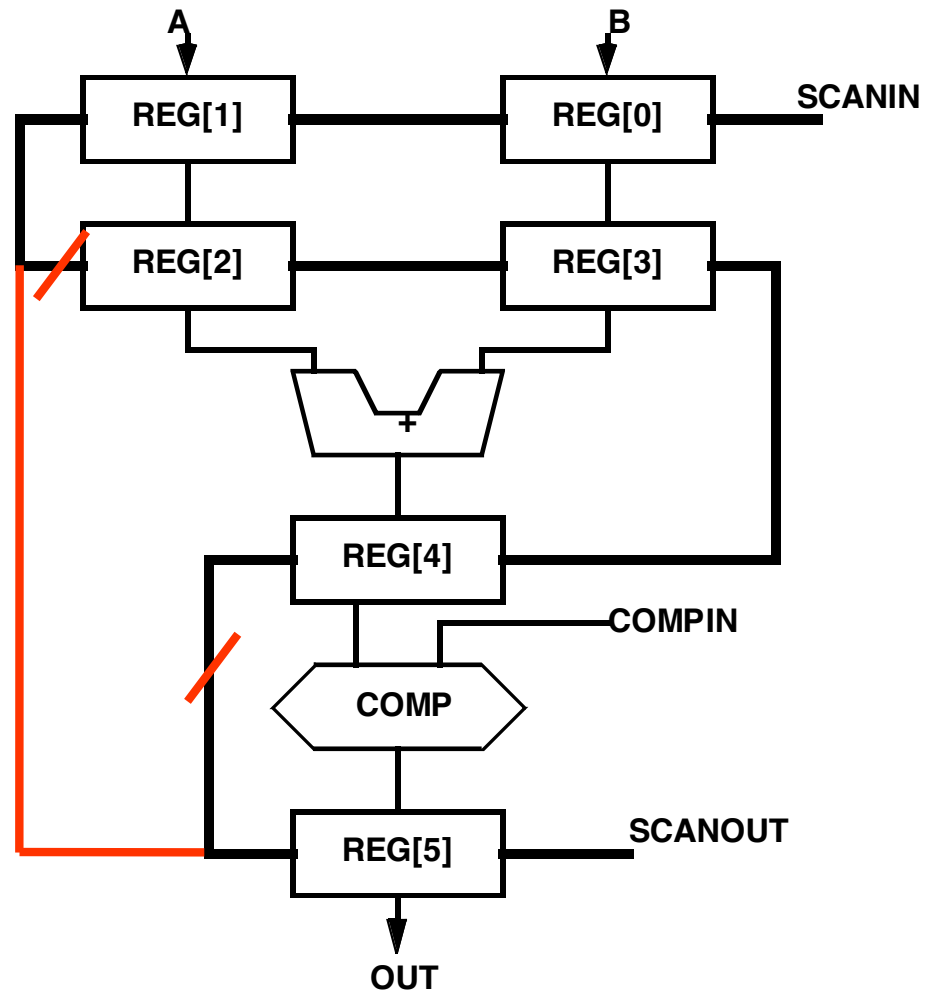


Scannable Flip-flops



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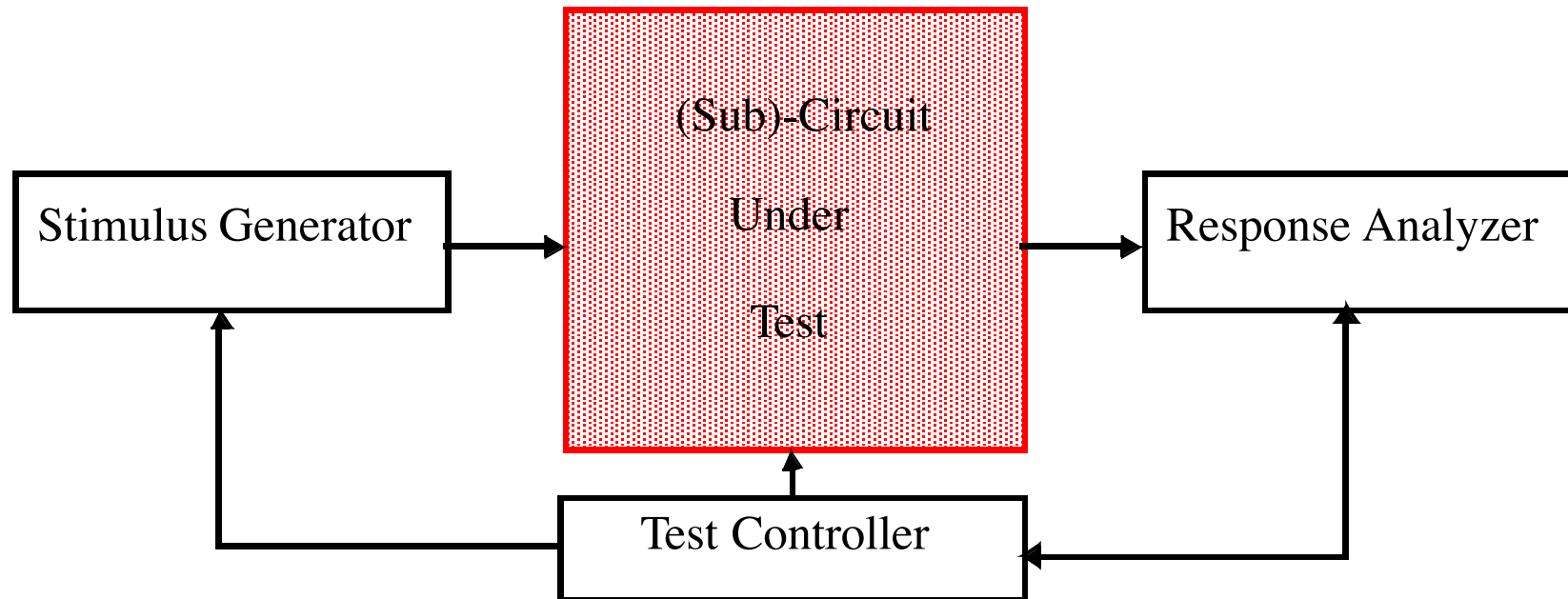
Scan-Path Testing



Partial-Scan can be more effective for pipelined datapaths

Built-in Self-test

- ◆ Built-in self-test lets blocks test themselves
 - Generate pseudo-random inputs to comb. logic
 - Combine outputs into a *syndrome*
 - With high probability, block is fault-free if it produces the expected syndrome



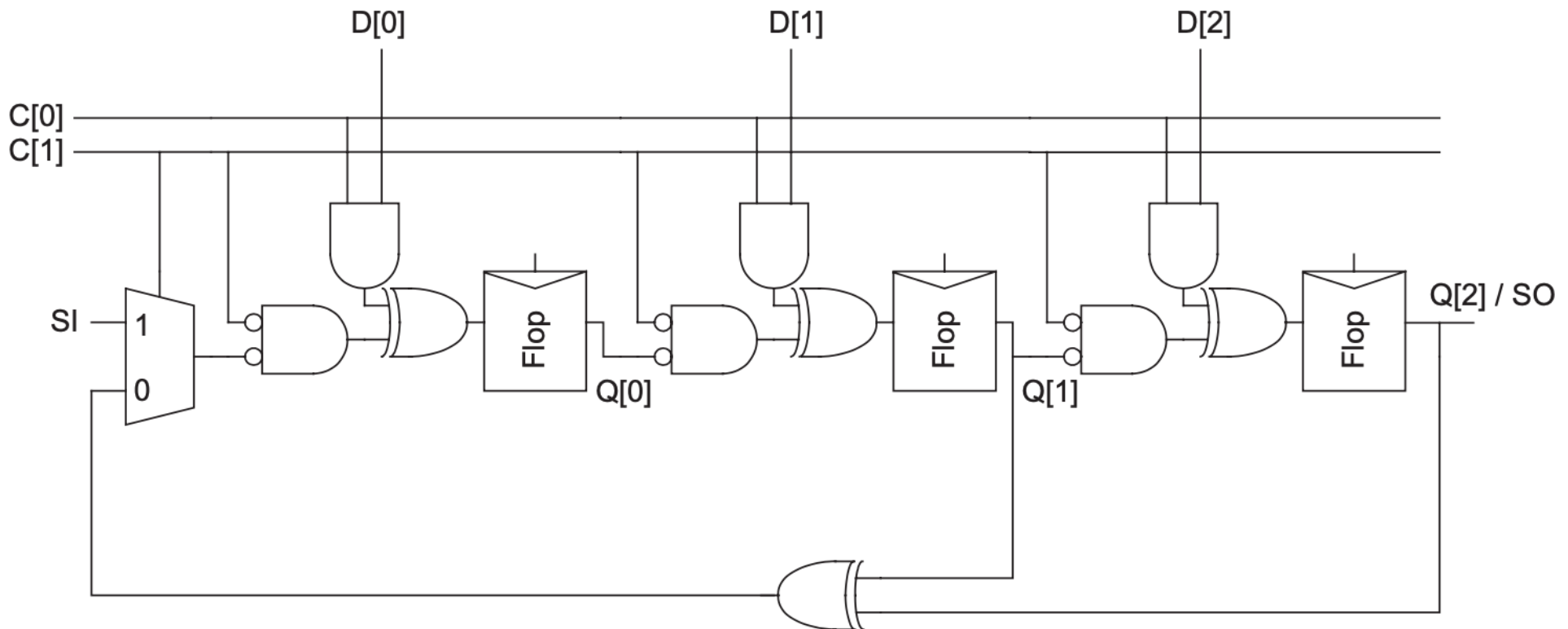
Rapidly becoming more important with increasing chip-complexity and larger modules

BILBO circuit

◆ Built-in Logic Block Observer (**BILBO**)

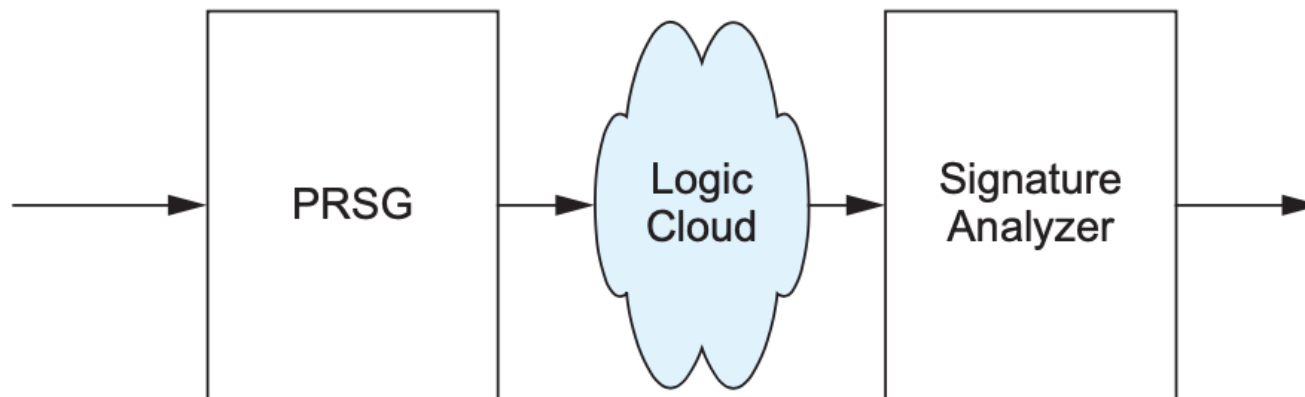
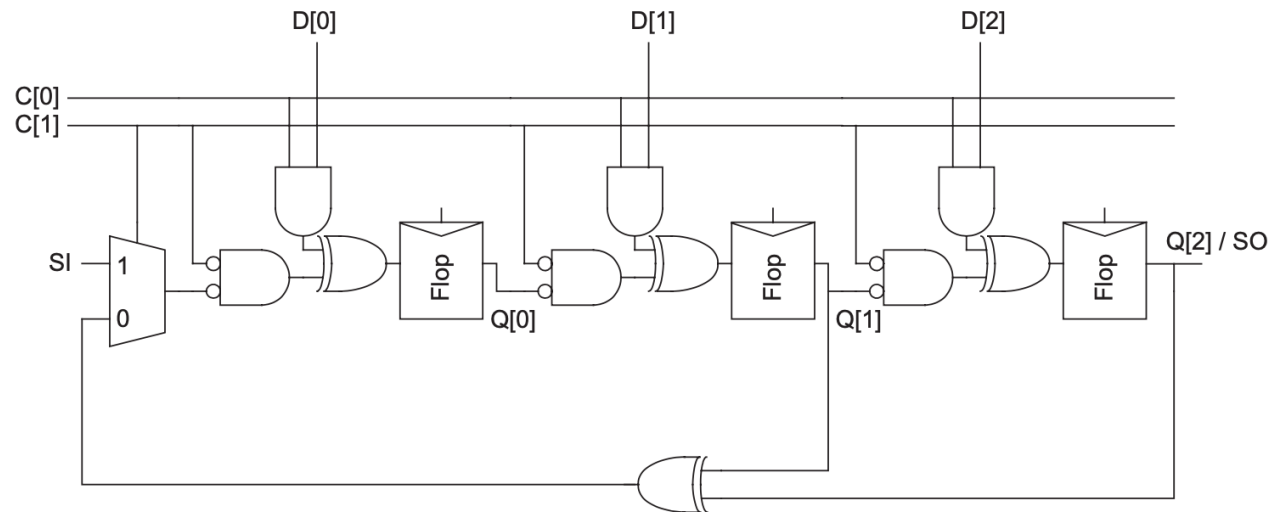
- Combine scan with PRSG & signature analysis

MODE	C[1]	C[0]
Scan	0	0
Test	0	1
Reset	1	0
Normal	1	1



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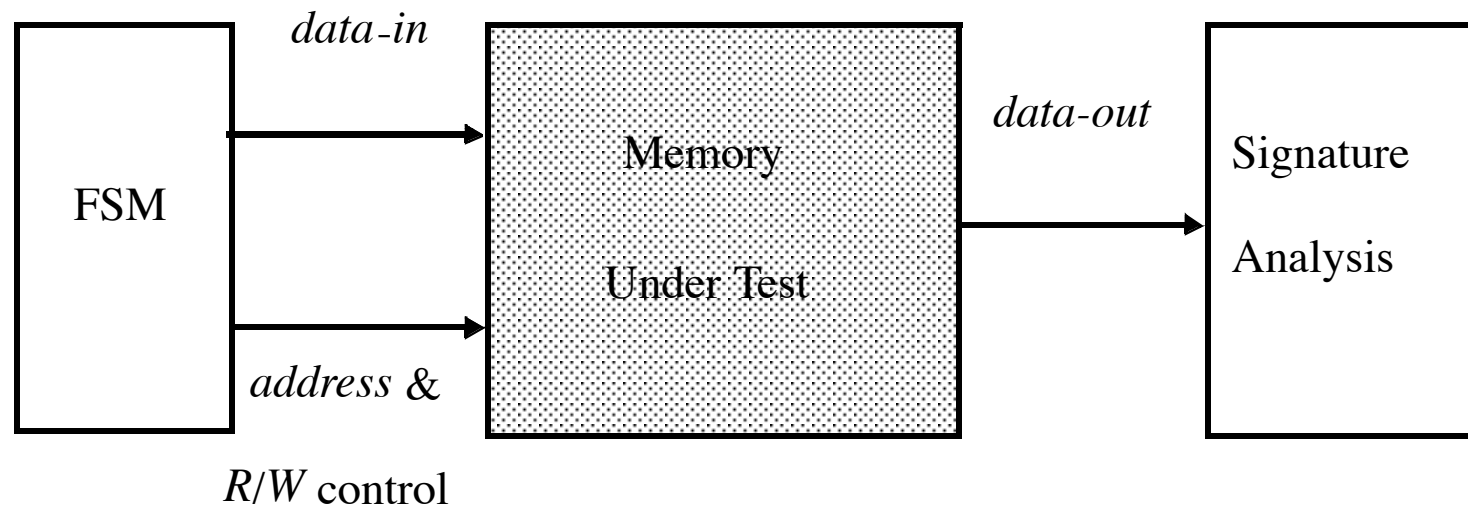
BILBO in Action



MODE	C[1]	C[0]
Scan	0	0
Test	0	1
Reset	1	0
Normal	1	1

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Memory Self-Test



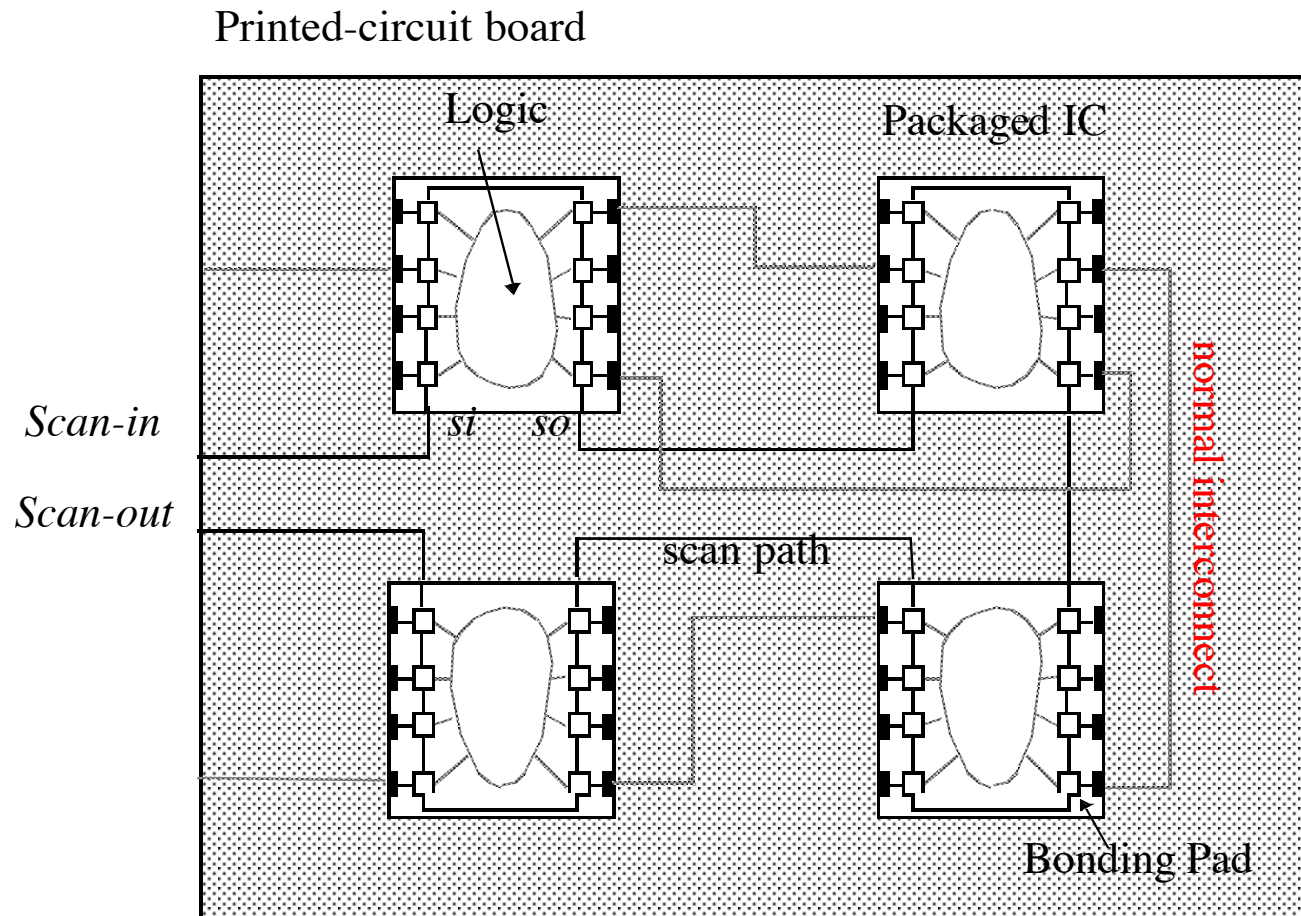
Common data test patterns

Pattern Name	Description	Purpose
All-Zeros (0x00)	Write all 0s, then read back	Detect stuck-at-1 faults
All-Ones (0xFF)	Write all 1s, then read back	Detect stuck-at-0 faults
Checkerboard (0xAA / 0x55)	Alternating 1s and 0s: 10101010 / 01010101	Detect coupling faults between adjacent cells
Walking 1 / Walking 0	Move a single '1' (or '0') across all bit positions	Detect bitline and wordline coupling, address decoder faults
Galloping	Move a '1' through memory, writing and reading surrounding cells	Detects cell-to-cell interference (coupling faults)
Random Data Pattern	Pseudorandom or LFSR-generated data	Detects pattern-sensitive faults

Board level testing – JTAG Boundary Scan

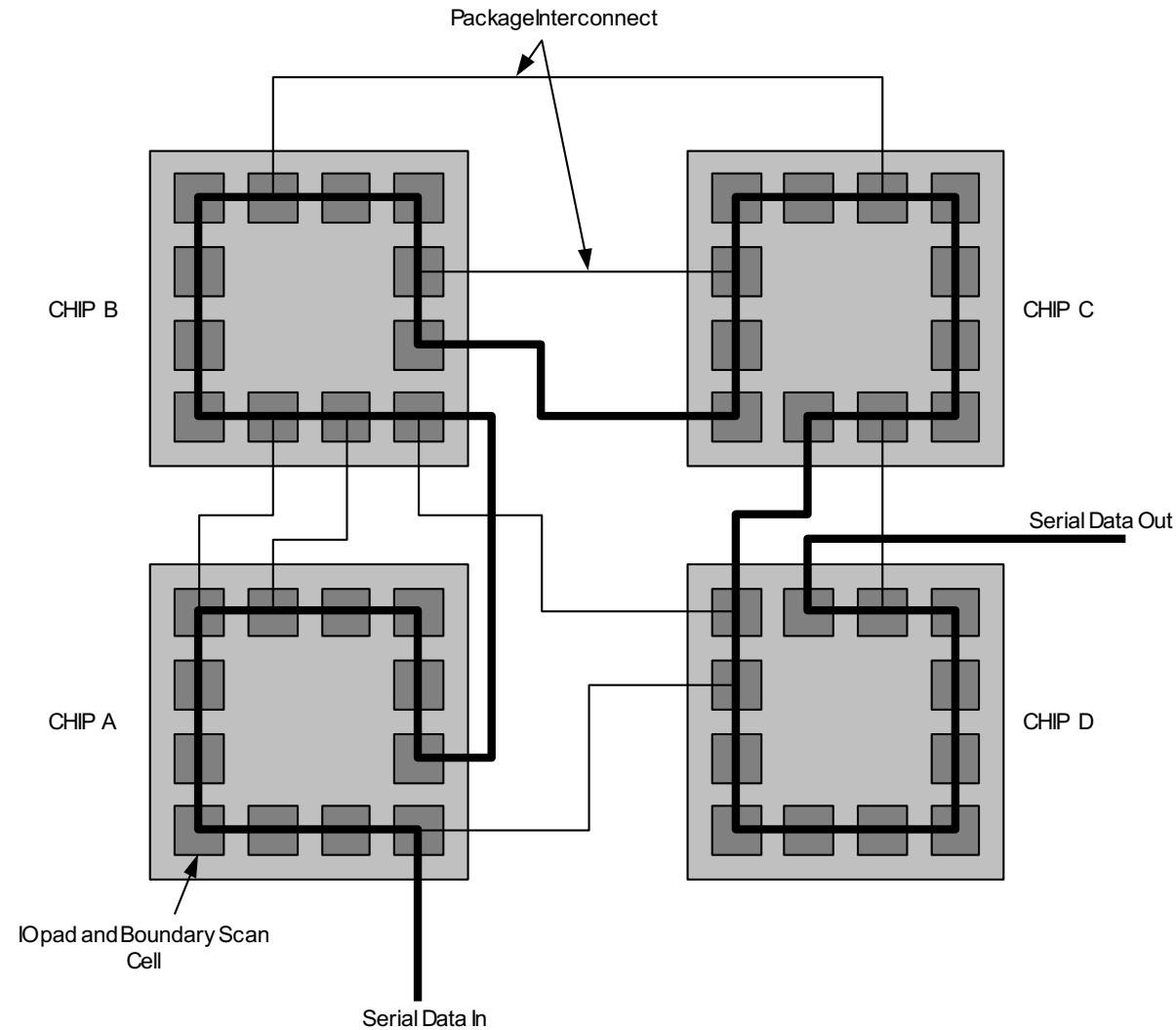
- ◆ Testing boards is also difficult
 - Need to verify solder joints are good
 - Drive a pin to 0, then to 1
 - Check that all connected pins get the values
- ◆ Through-hole boards used “bed of nails”
- ◆ SMT and BGA boards cannot easily contact pins
- ◆ Build capability of observing and controlling pins into each chip to make board test easier

JTAG Boundary Scan



Board testing becomes as problematic as chip testing

JTAG Boundary Scan Example



JTAG interface signals

- ◆ **Boundary scan is accessed through five pins**
 - **TCK:** test clock
 - **TMS:** test mode select
 - **TDI:** test data in
 - **TDO:** test data out
 - **TRST*:** test reset (optional)

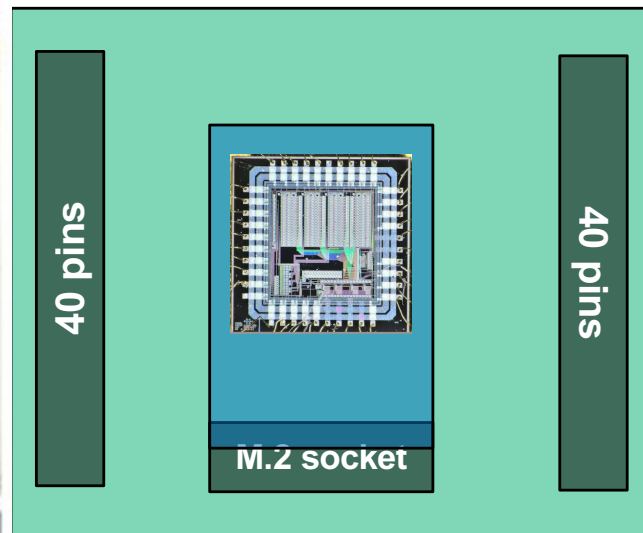
- ◆ **Chips with internal scan chains can access the chains through boundary scan for unified test strategy.**

Test Environment

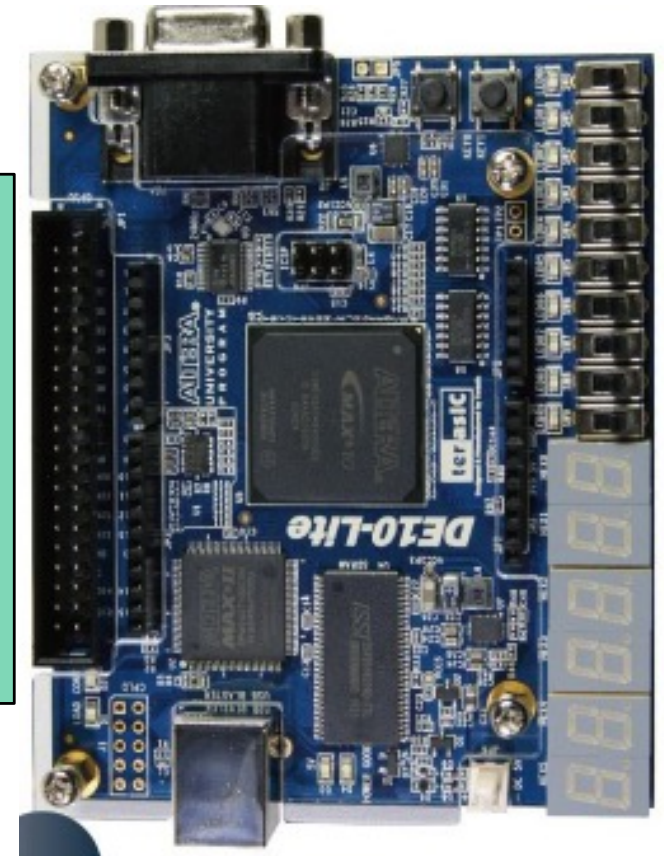
Test Driver Board



Test Board



Design Emulation Board



Summary

- ◆ **Think about testing from the beginning**
 - **Simulate as you go**
 - **Plan for test after fabrication**
- ◆ **“If you don’t test it, it won’t work! (Guaranteed)”**